10. Control chart for Mean
LCL = 4.774
CL = 4.982
UCL = 5.19

Control chart for Range
LCL = 0
CL = 0.36
UCL = 0.7614

## 10. Operations Research

### Exercise 10.1

5. $x_{11}=16,\ x_{12}=3, x_{22}=15,\ x_{23}=22, x_{33}=9,\ x_{34}=25$
Total Cost = ₹ 580

6. $x_{11}=30,\ x_{21}=5, x_{22}=28,\ x_{23}=7,\ x_{33}=25,\ x_{34}=25$
Total Cost = ₹ 1,076

7. $x_{11}=15,\ x_{13}=10, x_{23}=35,\ x_{31}=15,\ x_{32}=25,$
Total Cost = ₹ 580

8. $x_{11}=1,\ x_{12}=5, x_{24}=1,\ x_{31}=6,\ x_{33}=3,\ x_{34}=1,$
Total Cost = ₹ 102

9. $x_{11}=10,\ x_{13}=20, x_{21}=20,\ x_{22}=20,\ x_{24}=10,\ x_{32}=20$
Total Cost = ₹ 370

10. $x_{11}=3,\ x_{12}=1, x_{22}=2,\ x_{23}=4,\ x_{24}=2,\ x_{34}=3, x_{35}=6$
Total Cost = ₹ 153

11. (i) $x_{11}=7,\ x_{21}=3, x_{22}=9,\ x_{32}=1,\ x_{33}=10,$
Total Cost = ₹ 94

(ii) $x_{13}=7,\ x_{21}=10, x_{23}=2,\ x_{32}=10,\ x_{33}=1,$
Total Cost = ₹ 61

(iii) $x_{11}=7,\ x_{21}=2, x_{23}=10,\ x_{31}=1,\ x_{32}=10,$
Total Cost = ₹ 40

12. $x_{11}=200, x_{21}=50, x_{22}=175, x_{23}=125, x_{32}=150, x_{33}=250$
Total Cost = ₹ 12,200

### Exercise 10.2

4. 46 5. 280 6. 41 Hours 7. 37 8. 12

### Exercise 10.3

1. (i) $S_1$ (ii) $S_2$ 2. (a) Crop C (b) Crop B and Crop C
3. (i) Egg shampoo (ii) Egg Shampoo 4. (i) $A_3$ (ii) $A_2$ and $A_3$



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